PHP-FL class fairness

Complete results compendium

All committed result tables, including A/B/C, the five half-scale runs, the FedAPEN scope test, and the four-step diagnostic ladder.

Prepared 10 November 2026

Repository: github.com/Paul4805/phofl-class-fairness

Experiment inventory and status

Experiment	Status	Key result
Original Table-2 baseline (AM 66.94 / FM 8.07)	Not executed	No reproduction claim
A: fairness metrics (Fashion-MNIST smoke)	Complete	FM 0.1607; tail not tail-worse
B: aggregation reweighting (smoke)	Complete	Null (p 0.469-0.922)
C: correlated participation (smoke)	Complete	Pattern effect loses Holm
C pinned-stack smoke	Complete	Signs replicate; significance does not
C half-scale (4 runs)	Complete	FAIL pre-registered escalation
FedAPEN / linear half-scale	Complete	52.9-point gap; field-wide scope
Ladder 1-4 (half-scale)	Complete	Max 7.9% gap reduction
Seed-1 rerun	Not triggered	No step reached 30%
FedMRL / FedFV / FedHEAL / FedAWE / FedAU	Not run	Out of scope

All half-scale numbers use 10 clients, 60 rounds, one seed, four CNNs, one local epoch, 2,500 train / 500 test per client, and the pinned stack. Absolute AM/FM are not comparable to the paper's full-scale numbers.

A/B/C: Fashion-MNIST smoke results

A. Fairness metrics across completed smoke runs

exp	method	pattern	agg_mode	AM	FM	min_acc	p10	cvar10	jain	rho_p_acc
B	PHPFL	normal	inv_N	0.5105	0.1615	0.2600	0.2915	0.2600	0.9090	0.0667
B	PHPFL	normal	inv_p	0.5160	0.1628	0.2600	0.2780	0.2600	0.9095	-0.0182
B	PHPFL	normal	plain	0.5175	0.1607	0.2400	0.2985	0.2400	0.9120	-0.0061
C	PHPFL	entropy	plain	0.5120	0.1805	0.1750	0.2425	0.1750	0.8894	0.7212
C	PHPFL	linear	plain	0.5040	0.1439	0.2650	0.2875	0.2650	0.9246	-0.7576
C	Standalone	linear	none	0.4605	0.1856	0.1000	0.1990	0.1000	0.8603	-0.7939

A's original narrative (lowest FM but not best tail) was not confirmed: the FM winner also won min/CVaR, and rho was not positive.

B. Aggregation reweighting

agg_mode	AM	FM	min
plain	0.5175	0.1607	0.2400
inv_p	0.5160	0.1628	0.2600
inv_N	0.5105	0.1615	0.2600

comparison	statistic	p	median diff
plain vs inv_p [accuracy]	26.0000	0.9219	-0.0125
plain vs inv_N [accuracy]	16.0000	0.4688	0.0050
inv_p vs inv_N [accuracy]	17.0000	0.5703	0.0050

C. Correlated participation

pattern	method	AM	FM	min
entropy_correlated	PHPFL	0.5120	0.1805	0.1750
entropy_correlated	Standalone	0.4605	0.1856	0.1000
linear	PHPFL	0.5040	0.1439	0.2650
linear	Standalone	0.4605	0.1856	0.1000

C comparison	raw p	Holm p	loses sig
PHPFL vs Standalone [linear]	0.0645	0.1934	False
PHPFL vs Standalone [entropy_correlated]	0.0020	0.0098	False
linear vs entropy [accuracy]	0.4102	0.8203	False
linear vs entropy [dispersion]	0.0410	0.1641	True
linear vs entropy [advantage]	0.4102	0.8203	False

The dispersion effect lost Holm significance; PHP-FL vs Standalone under entropy survived. C is closed.

C escalation: pre-registered thresholds and outcomes

Quantity	Value	Rule
FM increase	-2.84%	GO over 25%; STOP under 10%
Min-accuracy change	-4.09 pts	GO over 5 drop; STOP under 2 drop
Rho difference	0.1697	GO over 1.0
Effect vs pinned smoke	-2.84% vs 15.42%	STOP if at most 1.1x pinned
Decision	FAIL	STOP / GO / MARGINAL

Pinned-stack smoke C

pattern	method	AM	FM	min	p10	CVaR10	rho
linear	PHPFL	0.5155	0.1511	0.2450	0.2990	0.2450	-0.7212
linear	Standalone	0.4575	0.1826	0.1050	0.1950	0.1050	-0.8424
entropy_correlated	PHPFL	0.5120	0.1744	0.2300	0.2570	0.2300	0.6930
entropy_correlated	Standalone	0.4575	0.1826	0.1050	0.1950	0.1050	0.6727

pinned comparison	raw p	Holm p	loses sig
PHPFL vs Standalone [linear]	0.0195	0.0781	True
PHPFL vs Standalone [entropy_correlated]	0.0020	0.0098	False
linear vs entropy [accuracy]	0.9219	1.0000	False
linear vs entropy [dispersion]	0.0840	0.2520	False
linear vs entropy [advantage]	0.8457	1.0000	False

The pinned stack preserves the sign but not the significance of the dispersion effect. PHP-FL vs Standalone under entropy remains significant; the pattern effect does not.

Five half-scale runs: two-axis fairness

Run	AM	client-FM	class_std	class gap	support gap	worst class	worst recall	best class	best recall
PHPFL/lin	0.6320	0.0785	0.1334	0.4430	0.5125	dog	0.2510	frog	0.6940
PHPFL/ent	0.6308	0.0763	0.1349	0.4560	0.4499	dog	0.2400	truck	0.6960
FedAPEN/lin	0.6209	0.0839	0.1698	0.5290	0.4159	cat	0.2560	frog	0.7850
Standalone/lin	0.5429	0.0962	0.1281	0.3860	0.3502	cat	0.2290	frog	0.6150
Standalone/ent	0.5467	0.0985	0.1196	0.3890	0.3541	cat	0.2180	frog	0.6070

Conservative headline: 44.3 points (PHP-FL/linear). Best client-FM run: PHP-FL/entropy at 45.6 points. FedAPEN: 52.9 points.

Per-class recall, full CIFAR-10 test set

run	airplane	automobile	bird	cat	deer	dog	frog	horse	ship	truck
PHPFL/lin	0.609	0.571	0.610	0.434	0.382	0.251	0.694	0.588	0.623	0.662
PHPFL/ent	0.626	0.613	0.599	0.457	0.397	0.240	0.673	0.511	0.616	0.696
FedAPEN/lin	0.520	0.704	0.631	0.256	0.482	0.275	0.785	0.639	0.663	0.424
Standalone/lin	0.405	0.586	0.564	0.229	0.335	0.285	0.615	0.381	0.488	0.541
Standalone/ent	0.451	0.579	0.531	0.218	0.369	0.306	0.607	0.398	0.476	0.554

Class gap = max - min of this row. Weak region: dog/cat/deer/horse.

Support filtering

run	total cells	supported	low support	threshold
PHPFL/lin	100	66	34	10
PHPFL/ent	100	66	34	10
Standalone/lin	100	66	34	10
Standalone/ent	100	66	34	10
FedAPEN/lin	100	66	34	10

170 low-support cells total; 45 zero-sample cells. Worst supported cell is 0.0 (client 0, cat, 11 samples).

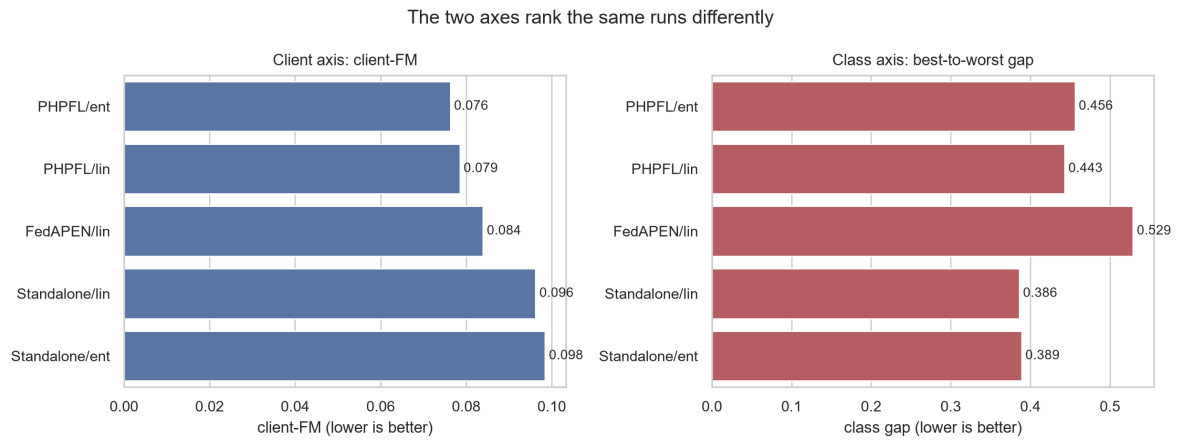
Ranking disagreement and AM normalization

run	client-FM	FM rank	class_std	std rank	class gap	gap rank
PHPFL/ent	0.0763	1	0.1349	4	0.4560	4
PHPFL/lin	0.0785	2	0.1334	3	0.4430	3
FedAPEN/lin	0.0839	3	0.1698	5	0.5290	5
Standalone/lin	0.0962	4	0.1281	2	0.3860	1
Standalone/ent	0.0985	5	0.1196	1	0.3890	2

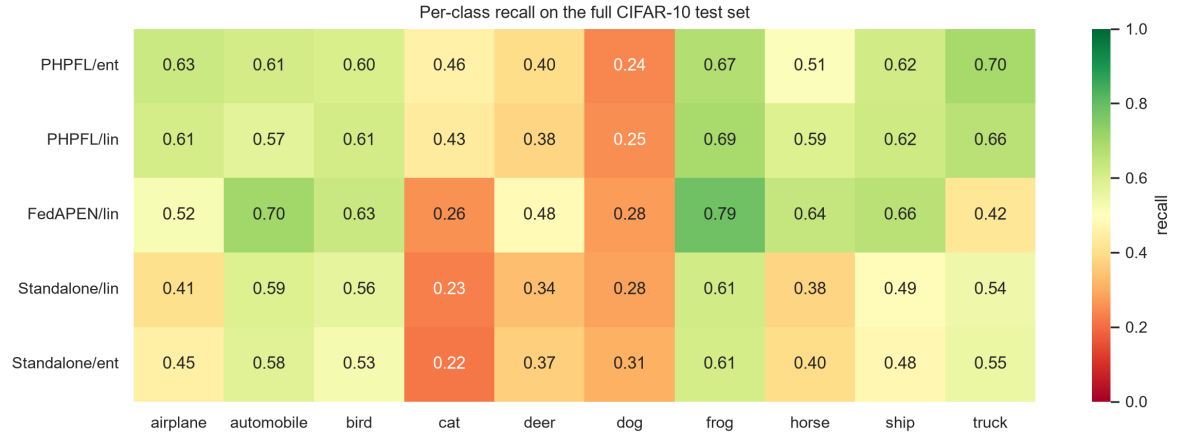
FedAPEN is 3rd by client-FM but 5th by class gap. Standalone/linear is 4th by client-FM but 1st by class gap. n=5, observation only.

run	AM	raw gap/AM	raw gap/ceil	support gap	support gap/AM	support gap/ceil
PHPFL/lin	0.6320	0.7010	0.5915	0.5125	0.8110	0.6803
PHPFL/ent	0.6308	0.7229	0.6000	0.4499	0.7132	0.6399
Standalone/lin	0.5429	0.7110	0.5006	0.3502	0.6450	0.4363
Standalone/ent	0.5467	0.7115	0.4974	0.3541	0.6477	0.4579
FedAPEN/lin	0.6209	0.8519	0.7110	0.4159	0.6697	0.5484

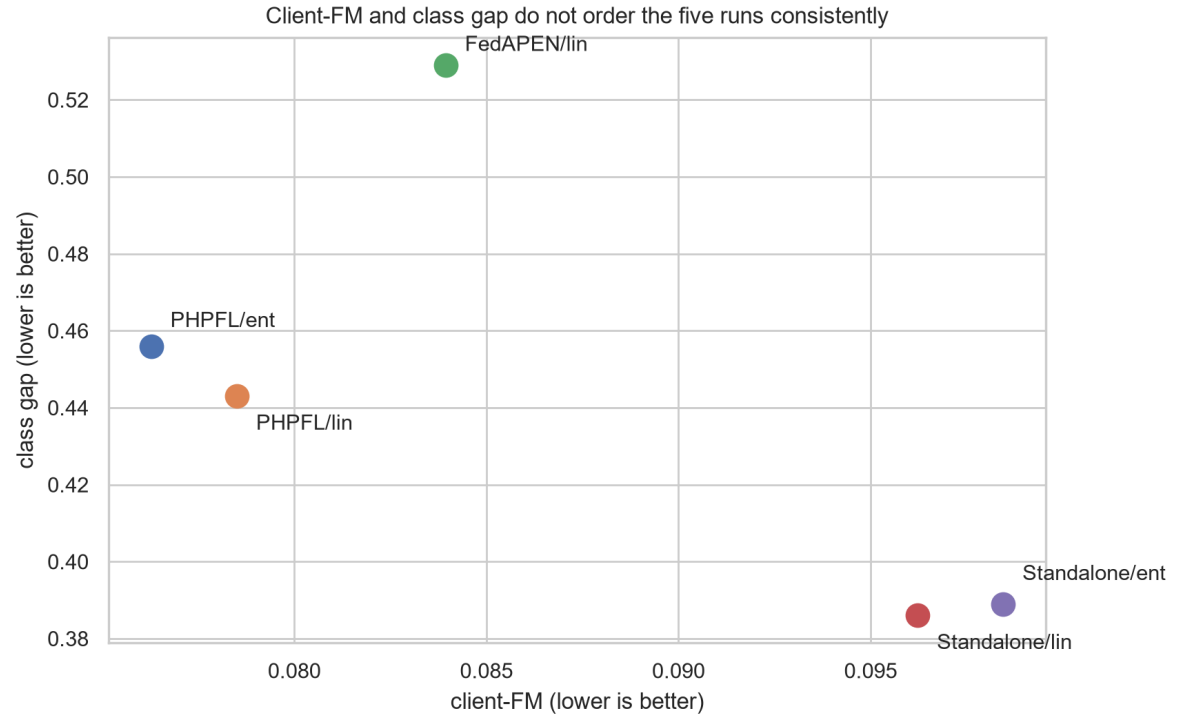
Standalone's raw gap advantage disappears after AM normalization (0.711 vs 0.701). The PHP-FL vs FedAPEN inversion survives (FedAPEN 3rd FM, 5th gap/AM).



Client-FM and class gap across the five runs.



Per-class recall matrix.

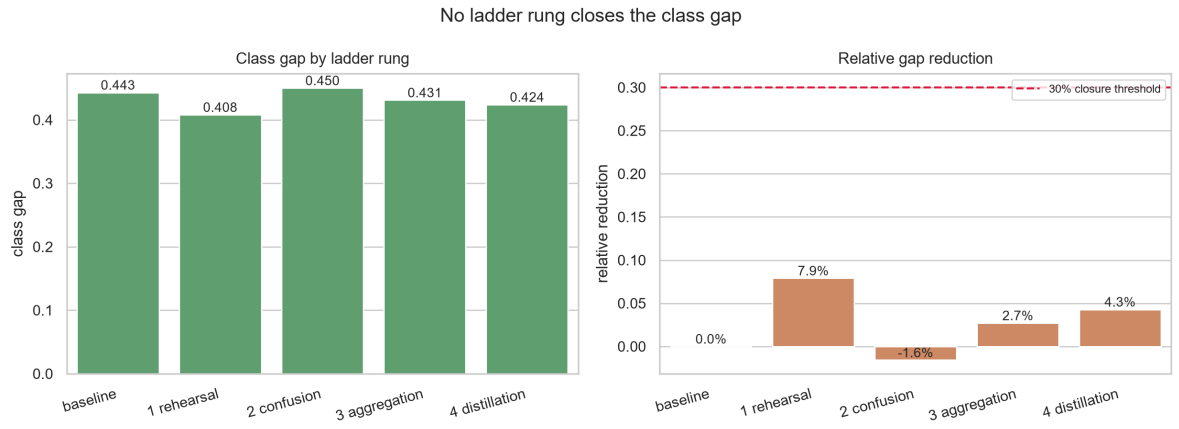


Client-FM versus class gap; the ordering is not monotone.

Diagnostic ladder: all four interventions

Rung	Intervention	AM	FM	gap	reduction	dog	frog	node
0	PHP-FL/linear baseline	0.6320	0.0785	0.443	-	0.251	0.694	-
1	class-balanced rehearsal	0.6099	0.0854	0.4080	+7.9%	0.3200	0.7280	dog 0.320
2	confusion-targeted loss	0.6346	0.0900	0.4500	-1.6%	0.2700	0.7200	dog 0.270
3	worst-class aggregation	0.6312	0.1002	0.4310	+2.7%	0.2920	0.7230	dog 0.292
4	per-class distillation	0.6322	0.0944	0.4240	+4.3%	0.2820	0.7060	dog 0.282

No rung reaches the 30% closure threshold; maximum reduction 7.9%. No seed-1 rerun was triggered.



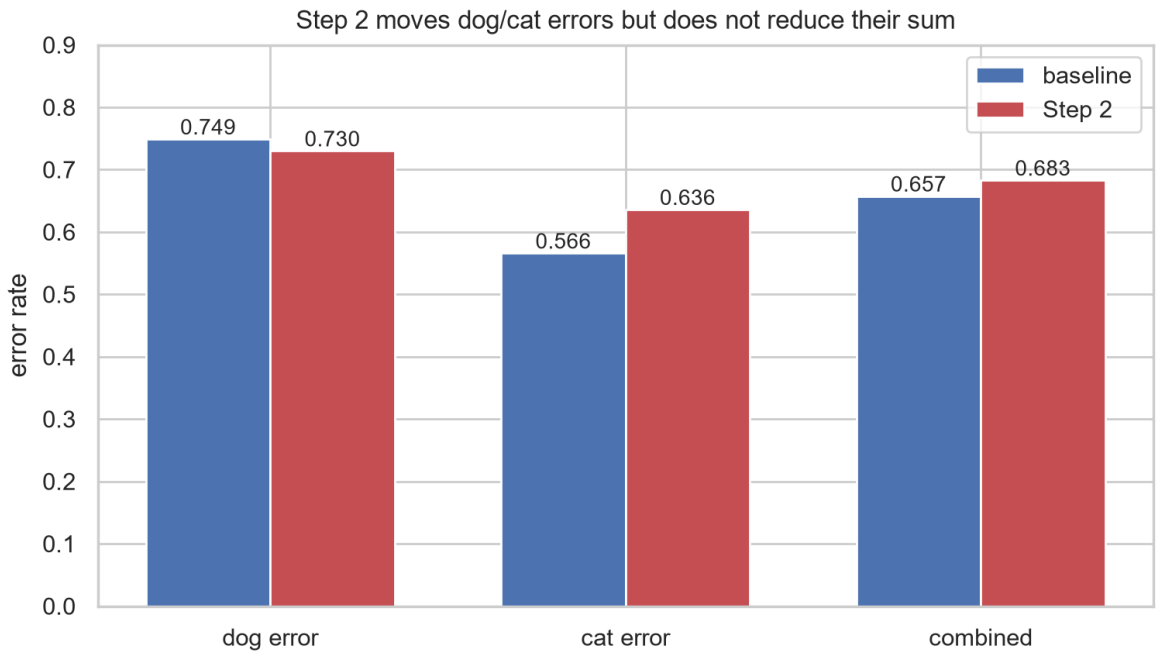
Ladder class gap and relative reduction.

Rung	Observed detail
1	dog 0.251 -> 0.320; frog 0.694 -> 0.728; gap 0.443 -> 0.408 (7.9%).
2	dog->cat 0.249 -> 0.191; cat recall 0.434 -> 0.364; combined dog+cat error 0.657 -> 0.683.
3	gamma=5.0; worst class airplane early, dog later; dog recall 0.292; gap 0.431 (2.7%).
4	T=1.0; best dog architecture EfficientNet-B1 0.599 -> 0.629, unchanged; gap 0.424 (4.3%).

Step-by-step per-class results

step	class	baseline	method	delta
1	airplane	+0.6090	+0.6060	-0.0030
1	automobile	+0.5710	+0.6570	+0.0860
1	bird	+0.6100	+0.4820	-0.1280
1	cat	+0.4340	+0.3770	-0.0570
1	deer	+0.3820	+0.5040	+0.1220
1	dog	+0.2510	+0.3200	+0.0690
1	frog	+0.6940	+0.7280	+0.0340
1	horse	+0.5880	+0.5690	-0.0190
1	ship	+0.6230	+0.6330	+0.0100
1	truck	+0.6620	+0.5910	-0.0710
2	airplane	+0.6090	+0.5960	-0.0130
2	automobile	+0.5710	+0.6050	+0.0340
2	bird	+0.6100	+0.6020	-0.0080
2	cat	+0.4340	+0.3640	-0.0700
2	deer	+0.3820	+0.4270	+0.0450
2	dog	+0.2510	+0.2700	+0.0190
2	frog	+0.6940	+0.7200	+0.0260
2	horse	+0.5880	+0.5670	-0.0210
2	ship	+0.6240	+0.6540	+0.0300
2	truck	+0.6620	+0.6840	+0.0220
3	airplane	+0.6090	+0.5780	-0.0310
3	automobile	+0.5710	+0.6170	+0.0460
3	bird	+0.6100	+0.5600	-0.0500
3	cat	+0.4340	+0.4070	-0.0270
3	deer	+0.3820	+0.4420	+0.0600
3	dog	+0.2510	+0.2920	+0.0410
3	frog	+0.6940	+0.7230	+0.0290
3	horse	+0.5880	+0.6200	+0.0320
3	ship	+0.6230	+0.6580	+0.0350
3	truck	+0.6620	+0.6400	-0.0220
4	airplane	+0.6090	+0.5830	-0.0260
4	automobile	+0.5710	+0.5670	-0.0040
4	bird	+0.6100	+0.5900	-0.0200
4	cat	+0.4340	+0.4570	+0.0230
4	deer	+0.3820	+0.3930	+0.0110
4	dog	+0.2510	+0.2820	+0.0310
4	frog	+0.6940	+0.7060	+0.0120
4	horse	+0.5880	+0.5790	-0.0090
4	ship	+0.6230	+0.6650	+0.0420
4	truck	+0.6620	+0.6660	+0.0040

Step 2 zero-sum: dog->cat 0.249 -> 0.191 (-23%); cat recall 0.434 -> 0.364 (-16%); combined dog+cat error 0.657 -> 0.683 (rose).



Step 2 shifts dog/cat errors without reducing their combined rate.

All statistical tests

Half-scale C tests (raw and Holm)

metric	comparison	test	n	stat	raw p	Holm p	Holm sig
accuracy	PHPFL vs Standalone [linear]	wilcoxon	10	0.0000	0.0020	0.0176	True
accuracy	PHPFL vs Standalone [entropy_correlated]	wilcoxon	10	0.0000	0.0020	0.0176	True
accuracy	linear vs entropy [accuracy]	wilcoxon	10	26.5000	1.0000	1.0000	False
|acc - group mean|	linear vs entropy [dispersion]	wilcoxon	10	27.0000	1.0000	1.0000	False
min-accuracy tail shortfall	linear vs entropy [min tail-shortfall]	wilcoxon	10	7.0000	0.8927	1.0000	False
CVaR10 tail shortfall	linear vs entropy [CVaR10 tail-shortfall]	wilcoxon	10	7.0000	0.8927	1.0000	False
min-accuracy	linear vs entropy [min]	paired bootstrap	10	-0.0256	0.2088	1.0000	False
CVaR10	linear vs entropy [CVaR10]	paired bootstrap	10	-0.0256	0.2088	1.0000	False
PHPFL - Standalone	linear vs entropy [advantage]	wilcoxon	10	26.5000	1.0000	1.0000	False

B smoke tests

metric	comparison	n	stat	p	median diff
accuracy	plain vs inv_p [accuracy]	10	26.0000	0.9219	-0.0125
accuracy	plain vs inv_N [accuracy]	10	16.0000	0.4688	0.0050
accuracy	inv_p vs inv_N [accuracy]	10	17.0000	0.5703	0.0050
accuracy	PHPFL vs Standalone [linear]	10	9.0000	0.0645	0.0325
accuracy	PHPFL vs Standalone [entropy_correlated]	10	0.0000	0.0020	0.0425
accuracy	linear vs entropy [accuracy]	10	15.0000	0.4102	-0.0125
|acc - group mean|	linear vs entropy [dispersion]	10	7.5000	0.0410	0.0225
PHPFL - Standalone	linear vs entropy [advantage]	10	15.0000	0.4102	0.0125

C smoke tests

metric	comparison	n	stat	p	median diff
accuracy	PHPFL vs Standalone [linear]	10	9.0000	0.0645	0.0325
accuracy	PHPFL vs Standalone [entropy_correlated]	10	0.0000	0.0020	0.0425
accuracy	linear vs entropy [accuracy]	10	15.0000	0.4102	-0.0125
|acc - group mean|	linear vs entropy [dispersion]	10	7.5000	0.0410	0.0225
PHPFL - Standalone	linear vs entropy [advantage]	10	15.0000	0.4102	0.0125

Appendix: per-client accuracies

Five half-scale runs

run	client	arch	acc	rare acc
PHPFL/lin	0	googlenet	0.5800	0.0000
PHPFL/lin	1	densenet121	0.6000	0.0000
PHPFL/lin	2	efficientnet-b1	0.6760	0.0000
PHPFL/lin	3	resnet18	0.7020	0.0000
PHPFL/lin	4	googlenet	0.6040	
PHPFL/lin	5	densenet121	0.7100	0.0000
PHPFL/lin	6	efficientnet-b1	0.5288	
PHPFL/lin	7	resnet18	0.7360	0.0000
PHPFL/lin	8	googlenet	0.6880	0.0000
PHPFL/lin	9	densenet121	0.4920	0.0000
PHPFL/ent	0	googlenet	0.5700	0.0000
PHPFL/ent	1	densenet121	0.5600	0.0000
PHPFL/ent	2	efficientnet-b1	0.6720	0.0000
PHPFL/ent	3	resnet18	0.7100	0.0000
PHPFL/ent	4	googlenet	0.6080	
PHPFL/ent	5	densenet121	0.6740	0.0000
PHPFL/ent	6	efficientnet-b1	0.5329	
PHPFL/ent	7	resnet18	0.7640	0.0000
PHPFL/ent	8	googlenet	0.6800	0.0000
PHPFL/ent	9	densenet121	0.5340	0.1429
Standalone/lin	0	googlenet	0.5240	0.0000
Standalone/lin	1	densenet121	0.4860	0.0000
Standalone/lin	2	efficientnet-b1	0.5900	0.0000
Standalone/lin	3	resnet18	0.6100	0.0000
Standalone/lin	4	googlenet	0.5400	
Standalone/lin	5	densenet121	0.5620	0.0000
Standalone/lin	6	efficientnet-b1	0.3683	
Standalone/lin	7	resnet18	0.6900	0.0000
Standalone/lin	8	googlenet	0.6480	0.0000
Standalone/lin	9	densenet121	0.4060	0.0000
Standalone/ent	0	googlenet	0.5260	0.0000
Standalone/ent	1	densenet121	0.4620	0.0000
Standalone/ent	2	efficientnet-b1	0.5960	0.0000
Standalone/ent	3	resnet18	0.6400	0.0000
Standalone/ent	4	googlenet	0.5380	
Standalone/ent	5	densenet121	0.5660	0.0000
Standalone/ent	6	efficientnet-b1	0.3992	
Standalone/ent	7	resnet18	0.6900	0.0000
Standalone/ent	8	googlenet	0.6540	0.0000
Standalone/ent	9	densenet121	0.3920	0.0000
FedAPEN/lin	0	googlenet	0.5900	0.0000
FedAPEN/lin	1	densenet121	0.5460	0.0000
FedAPEN/lin	2	efficientnet-b1	0.6720	0.0000
FedAPEN/lin	3	resnet18	0.6340	0.0000
FedAPEN/lin	4	googlenet	0.6120	
FedAPEN/lin	5	densenet121	0.6660	0.0000
FedAPEN/lin	6	efficientnet-b1	0.5021	
FedAPEN/lin	7	resnet18	0.7900	0.0000
FedAPEN/lin	8	googlenet	0.6840	0.0000
FedAPEN/lin	9	densenet121	0.5100	0.5000

Diagnostic ladder runs

run	client	arch	acc	rare acc
STEP1	0	googlenet	0.5460	0.3333
STEP1	1	densenet121	0.5660	0.0000
STEP1	2	efficientnet-b1	0.6120	0.0000
STEP1	3	resnet18	0.6980	0.0000
STEP1	4	googlenet	0.5760	
STEP1	5	densenet121	0.6760	0.0000
STEP1	6	efficientnet-b1	0.5432	
STEP1	7	resnet18	0.7560	0.0000
STEP1	8	googlenet	0.6700	0.0000
STEP1	9	densenet121	0.4540	0.2857
STEP2	0	googlenet	0.5660	0.0000
STEP2	1	densenet121	0.5900	0.0000
STEP2	2	efficientnet-b1	0.6960	0.0000
STEP2	3	resnet18	0.7260	0.0000
STEP2	4	googlenet	0.6220	
STEP2	5	densenet121	0.7240	0.0000
STEP2	6	efficientnet-b1	0.5370	
STEP2	7	resnet18	0.7380	0.0000
STEP2	8	googlenet	0.6860	0.0000
STEP2	9	densenet121	0.4580	0.2143
STEP3	0	googlenet	0.5740	0.0000
STEP3	1	densenet121	0.5780	0.0000
STEP3	2	efficientnet-b1	0.6800	0.0000
STEP3	3	resnet18	0.7320	0.0000
STEP3	4	googlenet	0.5920	
STEP3	5	densenet121	0.7480	0.0000
STEP3	6	efficientnet-b1	0.5062	
STEP3	7	resnet18	0.7460	0.0000
STEP3	8	googlenet	0.7020	0.0000
STEP3	9	densenet121	0.4500	0.0000
STEP4	0	googlenet	0.5580	0.0000
STEP4	1	densenet121	0.5900	0.0000
STEP4	2	efficientnet-b1	0.6940	0.0000
STEP4	3	resnet18	0.7160	0.0000
STEP4	4	googlenet	0.5960	
STEP4	5	densenet121	0.7480	0.0000
STEP4	6	efficientnet-b1	0.5288	
STEP4	7	resnet18	0.7500	0.0000
STEP4	8	googlenet	0.6780	0.0000
STEP4	9	densenet121	0.4600	0.0000

Full client x class matrices are in phofl/analysis/matrices and two_axis_client_class.csv.

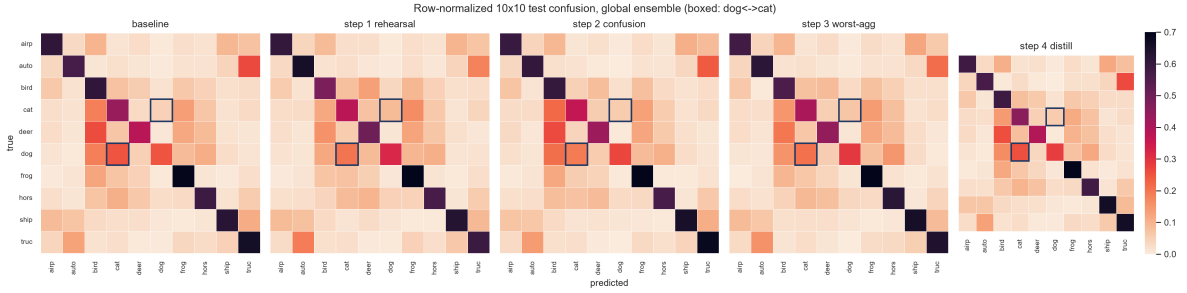
Confusion structure and pair error budgets

Full 10x10 matrices, global ensemble over 10 clients, full CIFAR-10 test set

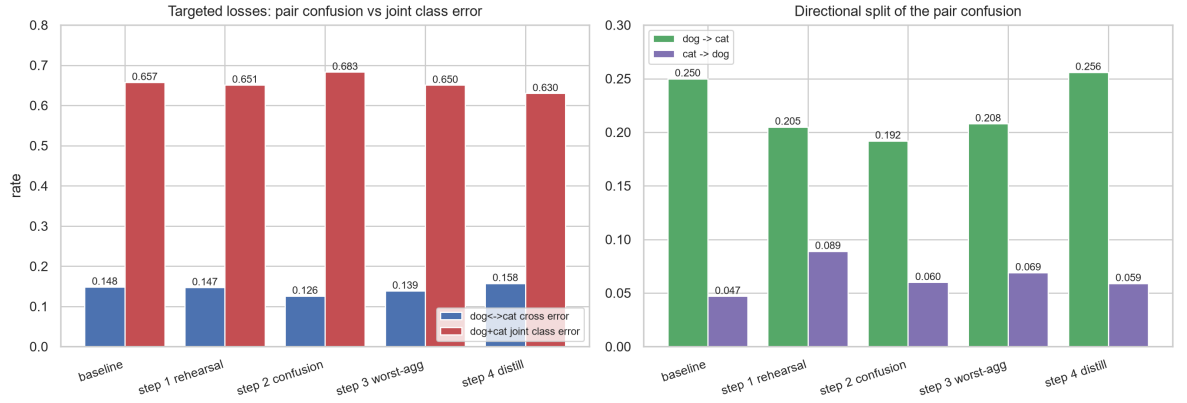
Cross error uses only the two off-diagonal cells; joint error is the average total error of the two classes, including errors to the other eight classes. Cross share is the pair's mutual mass over its total class error.

run	pair	a->b	b->a	cross	joint	cross share
baseline	dog-cat	0.2500	0.0470	0.1485	0.6575	0.2259
baseline	deer-horse	0.0900	0.0650	0.0775	0.5155	0.1503
baseline	automobile-truck	0.2660	0.1310	0.1985	0.3835	0.5176
STEP1_rehearsal	dog-cat	0.2050	0.0890	0.1470	0.6510	0.2258
STEP1_rehearsal	deer-horse	0.0710	0.0990	0.0850	0.4640	0.1832
STEP1_rehearsal	automobile-truck	0.1820	0.1910	0.1865	0.3760	0.4960
STEP2_confusion	dog-cat	0.1920	0.0600	0.1260	0.6830	0.1845
STEP2_confusion	deer-horse	0.0610	0.0850	0.0730	0.5030	0.1451
STEP2_confusion	automobile-truck	0.2430	0.1400	0.1915	0.3555	0.5387
STEP3_worst_agg	dog-cat	0.2080	0.0690	0.1385	0.6505	0.2129
STEP3_worst_agg	deer-horse	0.0870	0.0810	0.0840	0.4685	0.1793
STEP3_worst_agg	automobile-truck	0.2150	0.1580	0.1865	0.3710	0.5027
STEP4_distill	dog-cat	0.2560	0.0590	0.1575	0.6300	0.2500
STEP4_distill	deer-horse	0.0790	0.0740	0.0765	0.5140	0.1488
STEP4_distill	automobile-truck	0.2640	0.1320	0.1980	0.3835	0.5163

Step 2 lowers dog-cat cross error 15.2% but raises joint error 3.9%; dog->bird 0.188 -> 0.213 and cat->bird 0.187 -> 0.220 absorb the mass. Step 4 gives the lowest dog+cat joint error (0.6300) with cross error up 6.1%. Step 1 halves automobile/truck directional asymmetry (0.135 -> 0.009). Deer/horse is not a two-body pair: deer->bird alone is 0.261.



Row-normalised 10x10 test confusion for baseline and the four ladder rungs.

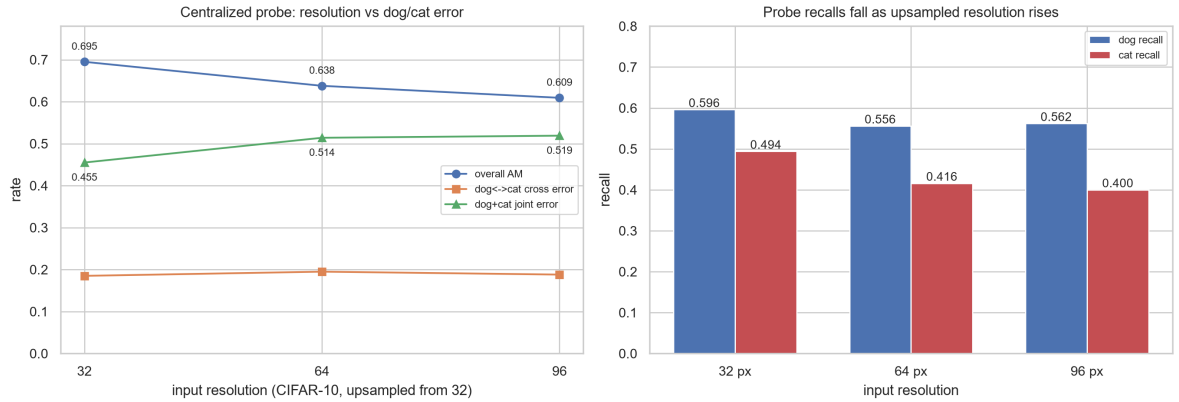


Dog/cat error budget: pair confusion falls under step 2 while joint class error rises.

Resolution probe (centralized)

2,000 train / 500 test images per class, one 3-conv CNN, 15 epochs, seed 0. CIFAR-10 is natively 32x32, so 64 and 96 are upsampled and add no information.

res	AM	dog	cat	dog+cat cross	dog+cat joint	deer+horse cross	auto+truck cross
32	0.6950	0.5960	0.4940	0.1850	0.4550	0.0730	0.0800
64	0.6378	0.5560	0.4160	0.1950	0.5140	0.0830	0.1200
96	0.6092	0.5620	0.4000	0.1880	0.5190	0.0870	0.1310



Upsampled resolution degrades the probe; pair cross-confusion stays flat.

Verdict

No confirmed resolution-limited floor and no confirmed label-ambiguity floor. The probe cannot test an information floor because 64/96 are upsampled, and it degraded as resolution rose. What the data does establish is that at the same native 32x32 input a centralized model reaches dog+cat joint error 0.455 against the federation's 0.6575, so the federated dog/cat deficit is not a missing-pixel problem. Native 96x96 STL-10 (downsampled to 64/32) is the required control. Targeted losses mainly move error between directions or out of the pair; the best joint improvement is 7.3% and no rung closes a pair.

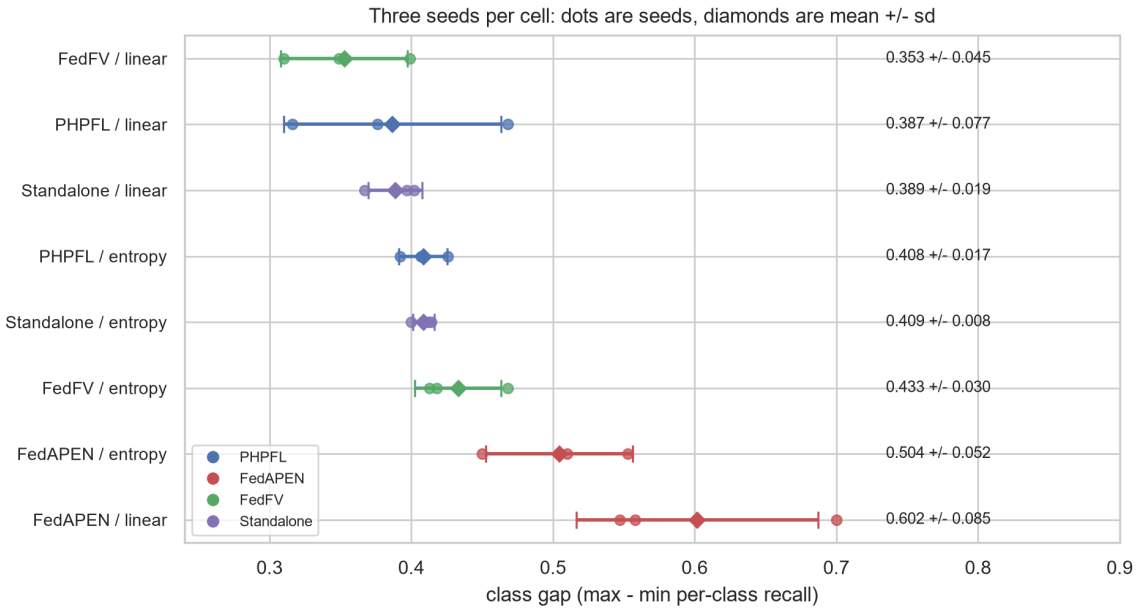
Three-seed class-gap sweep

Four methods x two patterns x three seeds, one stack (torch 2.14.1), 24 runs

Same protocol as the half-scale runs. Class gap is max minus min per-class recall on the full CIFAR-10 test set; error bars are sample sd over seeds. The seed-0 cells were re-run on the current stack, so the three seeds are comparable; the pinned seed-0 artifacts differ by at most 0.005 AM and 0.012 client-FM.

method / pattern	AM	client-FM	class gap	worst recall
PHPFL / entropy	0.6302 +/- 0.0041	0.0817 +/- 0.0032	0.4083 +/- 0.0170	0.278
PHPFL / linear	0.6286 +/- 0.0083	0.0860 +/- 0.0095	0.3867 +/- 0.0766	0.312
FedAPEN / entropy	0.6064 +/- 0.0075	0.0873 +/- 0.0109	0.5043 +/- 0.0517	0.276
FedAPEN / linear	0.6151 +/- 0.0030	0.0779 +/- 0.0137	0.6017 +/- 0.0853	0.207
FedFV / entropy	0.6262 +/- 0.0087	0.0847 +/- 0.0042	0.4330 +/- 0.0304	0.268
FedFV / linear	0.6272 +/- 0.0080	0.0874 +/- 0.0035	0.3527 +/- 0.0446	0.346
Standalone / entropy	0.5377 +/- 0.0060	0.0954 +/- 0.0100	0.4087 +/- 0.0076	0.212
Standalone / linear	0.5423 +/- 0.0014	0.0996 +/- 0.0095	0.3887 +/- 0.0189	0.215

The 44.3-point seed-0 PHP-FL figure is a high draw: the same cell spans 0.316-0.468. PHP-FL and Standalone are tied on the class axis (per-seed differences change sign). FedAPEN is worse than PHP-FL in 6/6 per-seed comparisons. FedFV/linear has the best mean gap and worst-class recall, but its 3.4-point margin is inside the seed noise (3/3 sign, p=0.25).



Per-seed class gap. Dots are seeds; diamonds are mean +/- sd.

Result artifact index

Committed result files

Path	Contents
phofl/results/A_fairness.csv	A fairness metrics
phofl/results/B_results.csv	B per-client aggregation results
phofl/results/C_results.csv	C per-client participation results
phofl/results/STATS_tests.csv	A/B/C Wilcoxon tests
phofl/results/C_holm.csv	C raw and Holm p-values
phofl/analysis/two_axis_run_summary.csv	Five-run two-axis summary
phofl/analysis/two_axis_fairness.csv	40-row client/class table
phofl/analysis/two_axis_client_class.csv	500-row client-class recall + support
phofl/analysis/two_axis_class_recall.csv	Global per-class recall
phofl/analysis/ranking_disagreement.csv	FM / class-gap ranks
phofl/analysis/am_normalization.csv	AM-normalized gaps
phofl/analysis/low_support_cells.csv	170 excluded cells
phofl_half/results/C_half_summary.csv	Five-run summary
phofl_half/results/C_half_stats_full.csv	Half-scale tests + Holm
phofl_half/results/C_half_results.csv	50 per-client rows
phofl_method/STEP*_summary.csv	Ladder per-step summary
phofl_method/STEP*_per_class.csv	Ladder per-class deltas
phofl_method/LADDER_SUMMARY.md	Ladder negative result
phofl/analysis/confusion_matrices/	10 raw + 10 normalised 10x10 matrices
phofl/analysis/confusion_summary.csv	Confusion AM and dog/cat recall
phofl/analysis/confusion_pair_budget.csv	Directional, cross, joint, share per pair
phofl/analysis/resolution_probe_cifar.csv	32/64/96 centralized probe
phofl/analysis/CONFUSION_RESOLUTION_FINDINGS.md	Confusion and resolution analysis
phofl_half/results/C_half_s3_seed_sweep.csv	24-run three-seed sweep
phofl/analysis/seed_sweep_summary.csv	Mean/sd per method x pattern
phofl/analysis/SEED_SWEEP_FINDINGS.md	Three-seed findings and FedFV notes

Excluded: checkpoints, run_store, generated data, third_party, phofl_pinned, .venv-pinned. See DATA.md.

No method contribution is claimed. The participation hypothesis is closed. class_FM is a reporting metric, not a training objective.